

Data Flow Diagram

Date	04 July 2026
Team ID	xxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

Data Flow Diagram Description

The Data Flow Diagram (DFD) illustrates how data flows through the **OptiCrop – Crop Recommendation System**. The process begins when the **Farmer/User** enters input values such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The system validates the entered data and retrieves the required information from the **Crop Dataset**. The validated data is then processed by the **Machine Learning Prediction Model**, which analyzes the input parameters and predicts the most suitable crop for cultivation. The recommended crop is displayed to the user, and the prediction details are

stored in the **Prediction History Database** for future reference. This data flow ensures accurate crop recommendations, efficient data management, and improved decision-making for farmers.

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